MACHINE LEARNING FINAL PROJECT

Online Purchase Intention Prediction

A Comparative Benchmark of 5 Supervised Classifiers

> How clickstream behavior, Google Analytics indicators, and class balancing strategies predict customer checkouts in e-commerce.

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DATASET & CODE

UCI Shoppers Intention (12.3k sessions)

Scikit-Learn Python Pipeline

The E-Commerce Conversion Dilemma

~15.5%

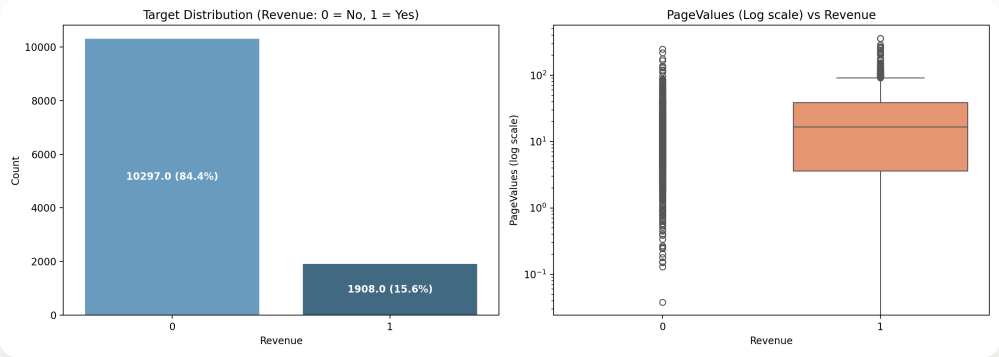
Actual purchasing sessions in dataset (84.5% leave without buying)

In online retail, millions browse but few buy. Intervening with discounts or live chat on every user causes huge profit leakage. Intervening on nobody loses high-value customers.

Goal:
Predict purchase intention in real-time so merchants can intervene only when it matters.

1. **The Accuracy Paradox**
   A naive model predicting "Nobody Buys" gets 84.5% accuracy but zero business value. We must optimize F1, Recall & PR-AUC.
2. **High Dimensional Sparsity**
   Combining numeric analytics with categorical sources (browser, OS, months) expands feature space to 68 dimensions after one-hot encoding.
3. **Five Fundamental Models Tested**
   Logistic Regression, KNN, Decision Tree, Random Forest, and SVM evaluated side-by-side.

UCI Dataset & Feature Architecture



Left: 84.5% vs 15.5% Class Imbalance | Right: Log-scale distribution of PageValues

12,205 Clean Unique Sessions

10 Numerical Behavioral Features:

- Page counts & browsing durations: `Administrative`, `Informational`, `ProductRelated`.
- Web analytics: `BounceRates`, `ExitRates`, `PageValues`, `SpecialDay`.

7 Categorical Session Contexts:

- User context: `VisitorType`, `OperatingSystems`, `Browser`, `Region`.
- Temporal cues: `Month`, `Weekend`, `TrafficType`.

> **Key Domain Finding:**
> PageValues is non-zero almost exclusively for users who complete checkout.

Leakage-Free Preprocessing Pipeline

A **Data Cleansing**

Removed 125 exact duplicate sessions. Zero missing values detected across all 18 columns.

B **Stratified 80/20 Holdout Split**

9,764 training sessions vs 2,441 test sessions. Preserved exact 15.5% positive ratio in both partitions.

C **ColumnTransformer Integration**

Numerical scaled via `StandardScaler`; Categorical transformed via `OneHotEncoder(handle_unknown='ignore')`.

D **Class Imbalance Compensation**

Employed inverse class frequency penalties inside loss functions: $w_j = \frac{N}{2 \times N_j}$. Prevents models from ignoring the rare positive class.

E **Stratified 5-Fold Cross Validation**

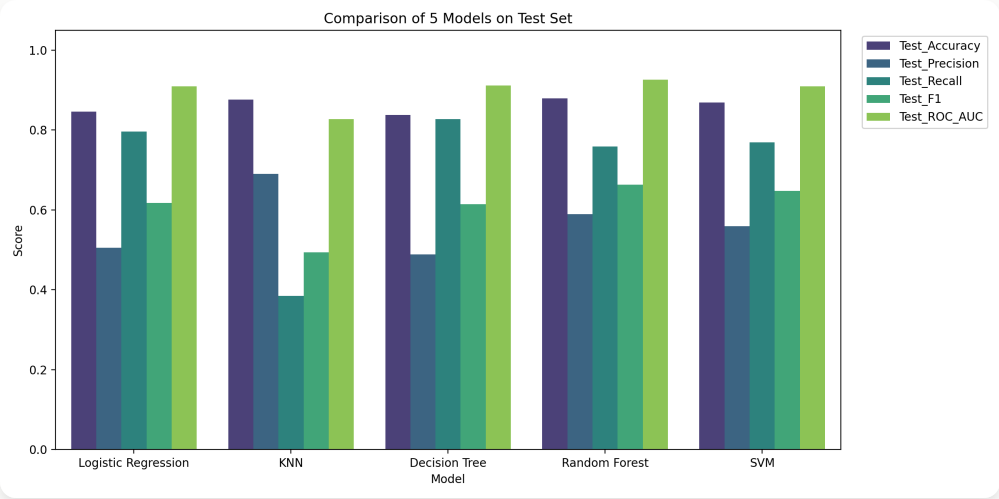
All hyperparameter stability tested with 5-fold CV before evaluating against holdout test data.

Benchmark Results on Holdout Test Set

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Logistic Regression	84.60%	0.5050	0.7958	0.6179	0.9097	0.6627
K-Nearest Neighbors (k=7)	87.67%	**0.6901**	0.3848	0.4941	0.8270	0.5717
Decision Tree (CART, d=6)	83.78%	0.4892	**0.8272**	0.6148	0.9119	0.6673
★ Random Forest (150 trees)	**87.96%**	**0.5894**	**0.7592**	**0.6636**	**0.9258**	**0.7096**
SVM (RBF Kernel)	86.89%	0.5589	0.7696	0.6476	0.9095	0.6695

Key Insight: Random Forest wins across F1–Score (0.6636), ROC-AUC (0.9258), and PR-AUC (0.7096).

Multi-Metric Performance Overview



✓ **Random Forest: The Balanced Leader**

Strikes the ideal sweet spot: high recall (75.9%) without collapsing precision (58.9%).

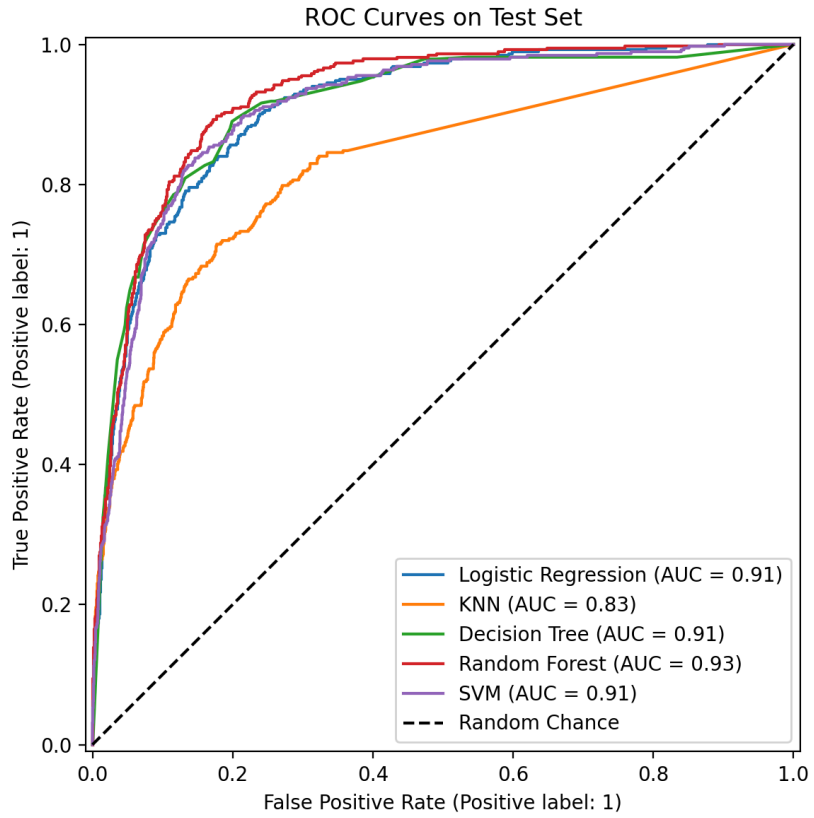
✗ **KNN: The Dimensionality Victim**

Highest precision (69.0%) but catastrophic recall (38.5%). Misses over 61% of buyers because Euclidean distance breaks down in 68-dimension sparse space.

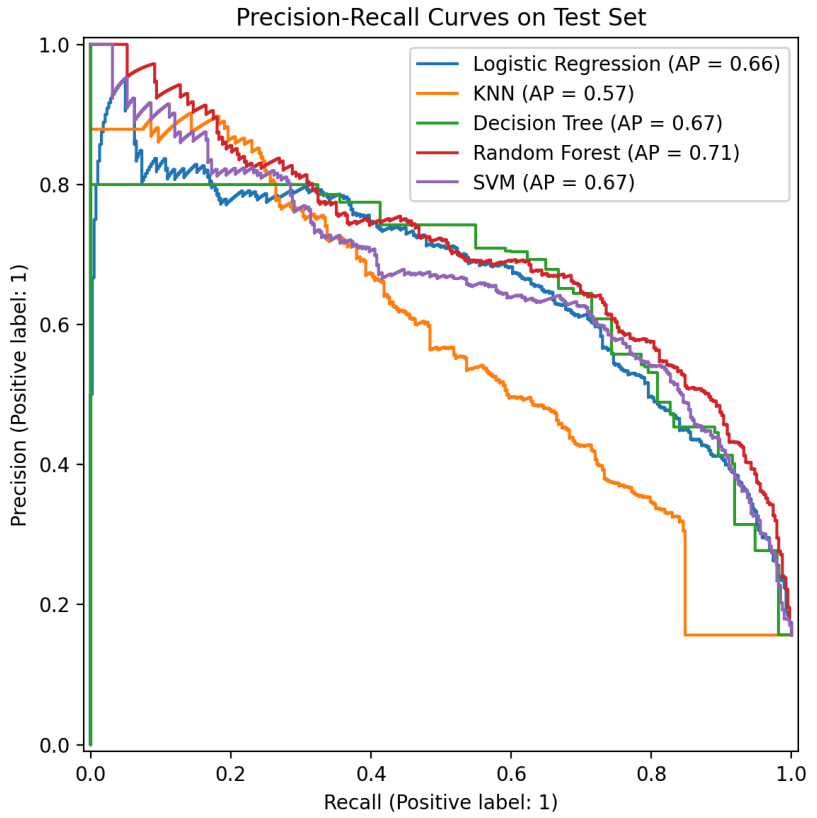
i **Decision Tree: Maximum Sensitivity**

Highest raw recall (82.7%) but lowest precision (48.9%). High false alarm rate.

ROC & Precision–Recall Trajectories

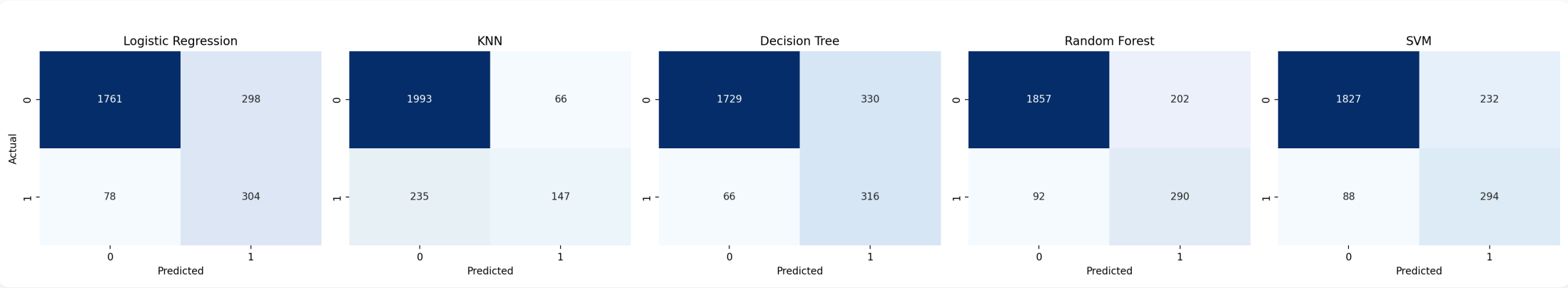


ROC Curves: Random Forest leads with AUC = 0.93, followed by Decision Tree and SVM.



PR Curves (Sensitive to Imbalance): Random Forest maintains superior precision across high recall thresholds (AP = 0.71).

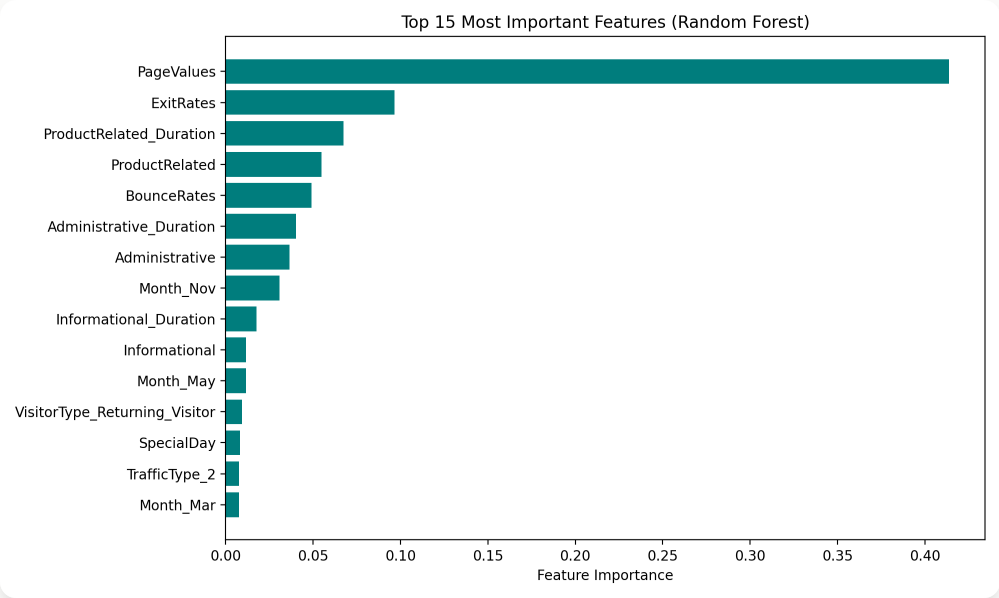
Operational Trade-Offs in Confusion Matrices



Decision Tree & Logistic: High False Positives (~250–310). Good when missing a customer is catastrophic, but sends too many unnecessary discounts.

Random Forest: Capped False Positives at 202 while capturing 290 out of 382 true buyers. Lowest overall commercial cost.

What Actually Drives the Purchase?



Top Predictors Decoded

1. **PageValues (Dominant >38%)**
   Users visiting pages with historical monetary association exhibit overwhelming conversion probability.

2. **ExitRates & BounceRates**
   Strong negative indicators. High bounce rates signal casual window shoppers.

3. **Month of November (Month_Nov)**
   Highest categorical rank. Direct reflection of Black Friday and holiday promotional rushes.

Conclusion & Business Recommendations

Key Takeaways

- **Best Model:** Random Forest is decisively superior for tabular e-commerce clickstreams (F1: 0.6636, ROC-AUC: 0.9258).
- **Tree Ensembles > Metric Models:** Bagging trees effortlessly absorb non-linear combinations and one-hot sparsity that crippled KNN.
- **Class Balancing is Mandatory:** Unweighted models fail to generate actionable business alerts.

Next Steps & Extensions

Production Deployment:

Deploy Random Forest as a streaming inference microservice (sub-50ms latency) triggered when PageValues > 0.

Future Algorithms:

Benchmark XGBoost, LightGBM, and CatBoost with Bayesian Hyperparameter Tuning.

Thank you! Q & A Session.